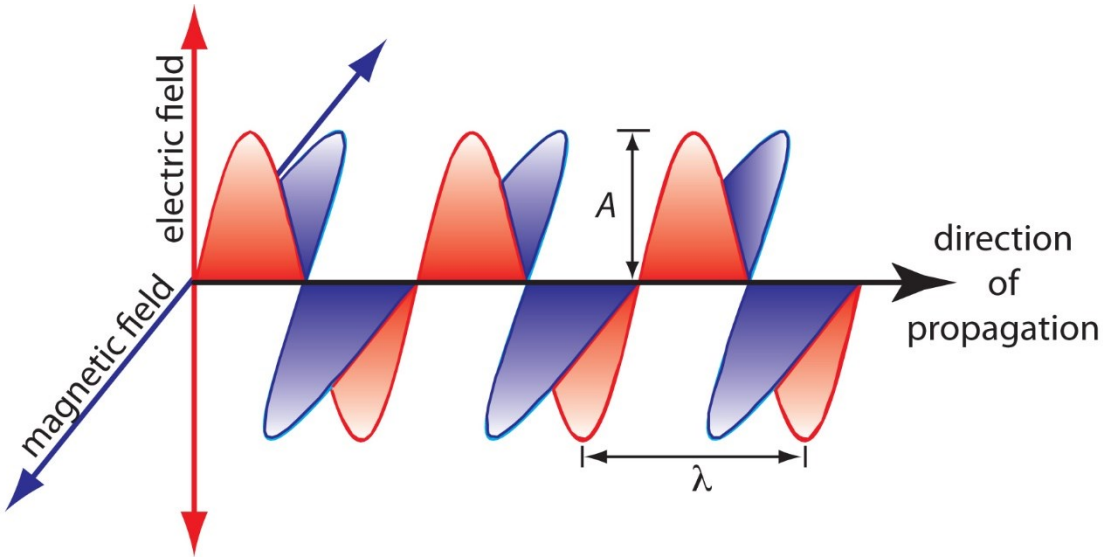

electric field
magnetic field
A
direction
of
propagation
λ

Plane Waves

Maxwell's Equations

In a source free lossless medium, $\mathbf{J} = \rho = \sigma = 0$.

Maxwell's equations:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t}$$

$$\varepsilon \nabla \cdot \mathbf{E} = 0$$

$$\mu \nabla \cdot \mathbf{H} = 0$$

Wave Equation

Take the curl of the first equation and make use of the second and the third equations, we have:

Note :

$$\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

$$\nabla^2 \mathbf{E} = \mu \frac{\partial}{\partial t} \nabla \times \mathbf{H} = \mu\varepsilon \frac{\partial^2}{\partial t^2} \mathbf{E}$$

This is called the **wave equation**:

$$\nabla^2 \mathbf{E} - \mu\varepsilon \frac{\partial^2}{\partial t^2} \mathbf{E} = 0$$

A similar equation for **H** can be obtained:

$$\nabla^2 \mathbf{H} - \mu\varepsilon \frac{\partial^2}{\partial t^2} \mathbf{H} = 0$$

Wave Equation (Free Space)

In free space, the wave equation for **E** is:

$$\nabla^2 \mathbf{E} - \mu_0 \varepsilon_0 \frac{\partial^2}{\partial t^2} \mathbf{E} = 0$$

where

$$\mu_0 \varepsilon_0 = \frac{1}{c^2}$$

c being the speed of light in free space ($\sim 3 \times 10^8$ (m/s)). Hence the speed of light can be derived from Maxwell's equation.

Maxwell's Equations for Time-Harmonic Fields

In phasor form, Maxwell's equations can be written as:

$$\nabla \times \mathbf{E} = -j\omega\mathbf{B}$$

$$\nabla \times \mathbf{H} = j\omega\mathbf{D}$$

$$\nabla \cdot \mathbf{D} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

Using the phasor form expression, the wave equation for **E** field is also called the **Helmholtz's equation**, which is:

$$\nabla^2\mathbf{E} + \mu_0\varepsilon_0\omega^2\mathbf{E} = \nabla^2\mathbf{E} + k^2\mathbf{E} = 0$$

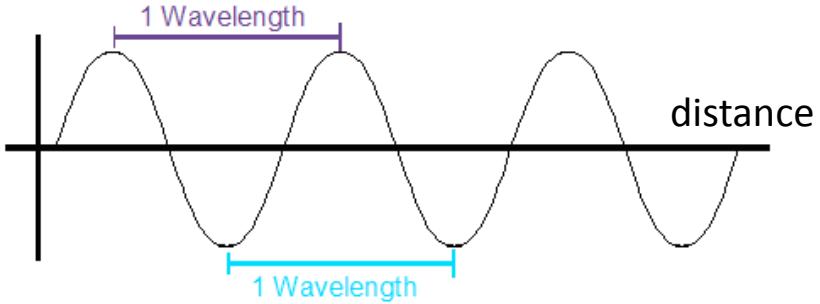
$$\text{where} \quad k = \omega\sqrt{\mu_0\varepsilon_0}$$

Propagation Constant (Wave Number)

k is called the **wavenumber** or the **propagation constant**.

$$k = k_0 = \omega\sqrt{\mu_0\varepsilon_0} = \frac{2\pi f}{c} = \frac{2\pi}{\lambda_0}$$

where λ_0 is the free space wavelength. $\lambda_0 = c / f$



Scalar Helmholtz's Equation

In Cartesian coordinates, the Helmholtz's equation can be written as three scalar equations in terms of the respective x, y, and z components of the **E** field. For example, the scalar equation for the E_x component is:

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}+k^2\right)E_x=0$$

Consider a special case of the E_x in which there is no variation of E_x in the x and y directions, i.e.,

$$\frac{\partial^2}{\partial x^2}E_x=\frac{\partial^2}{\partial y^2}E_x=0$$

Plane Wave Equation

This is called the plane wave condition and $E_x(z)$ now varies with z only. The wave equation for E_x becomes:

$$\frac{d^2E_x(z)}{dz^2} + k^2E_x(z) = 0$$

Note that a plane wave is not physically realizable because it extends to an infinite extent in the x and y directions. However, when considered over a small plane area, its propagation characteristic is very close to a spherical wave, which is a real and common form of electromagnetic wave propagating.

Plane Wave Solution

Solutions to the plane wave equation take one form of the following functions, depending on the boundary conditions:

1. $E_x(z) = E_0^+ e^{-jkz}$
2. $E_x(z) = E_0^- e^{+jkz}$
3. $E_x(z) = E_0^+ e^{-jkz} + E_0^- e^{+jkz}$

E_0^+ and E_0^- are constants to be determined by boundary conditions.

$E_0^+ e^{-jkz}$ and $E_0^- e^{+jkz}$ are plane waves propagating along the $+z$ direction and $-z$ direction.

Plane Wave Solution (cont.)

The accompanying magnetic field **H** can be found from Maxwell's equation

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$$

For example, if the solution for **E** is,

$$E_x(z) = E_0^+ e^{-jkz},$$

then the solution for **H** is:

$$H_y(z) = \frac{1}{-j\omega\mu}\frac{\partial E_x(z)}{\partial z} = \frac{k}{\omega\mu}E_0^+ e^{-jkz}$$

Plane Wave Solution (cont.)

The ratio of E_x to H_y is called the **intrinsic impedance** of the medium, η.

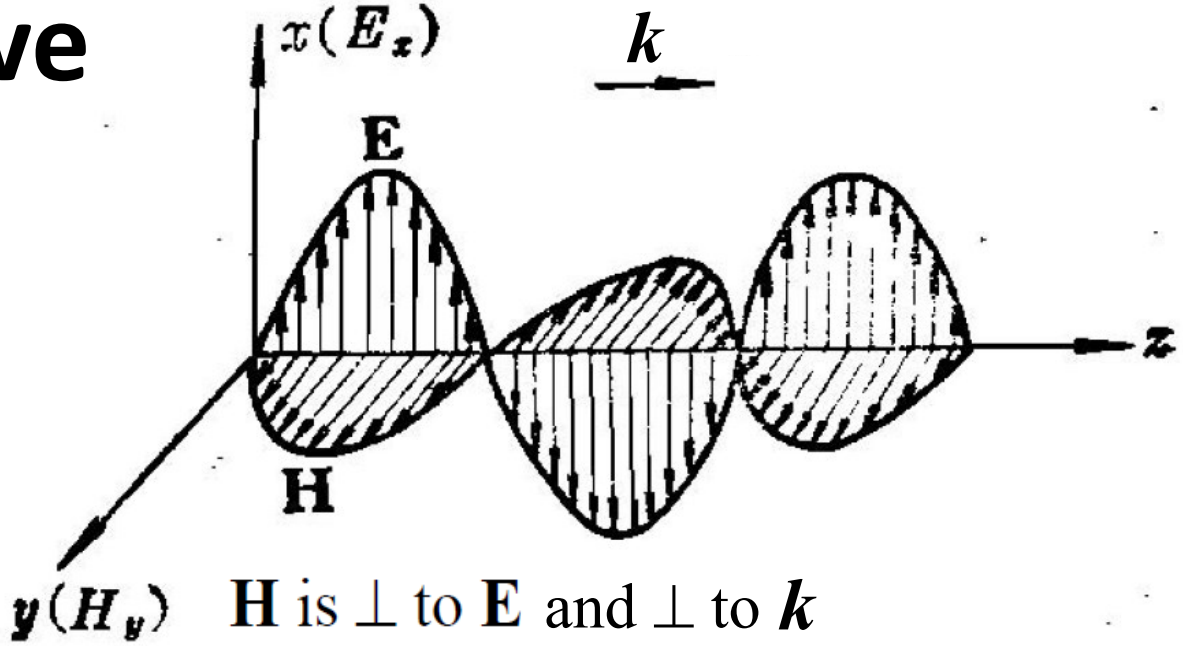
$$\eta = \frac{E_x(z)}{H_y(z)} = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\varepsilon}} = \sqrt{\frac{\mu}{\varepsilon}} \ (\Omega)$$

Note that η is independent of z. In free space,

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi \approx 377\ \Omega$$

and E_x and H_y are in phase (as η_0 is a real number).

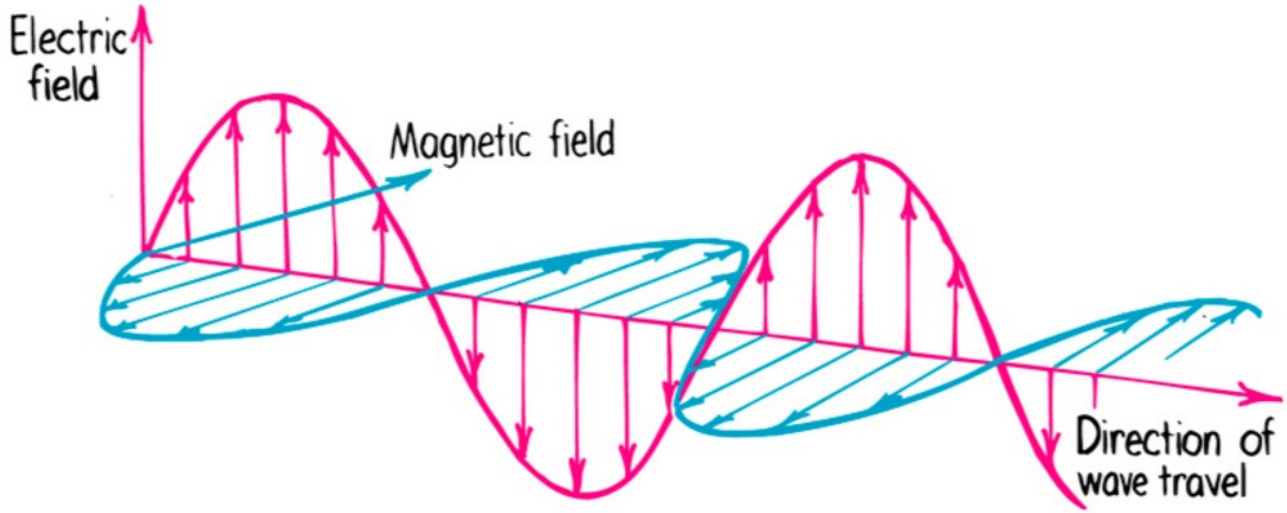
Plane Wave



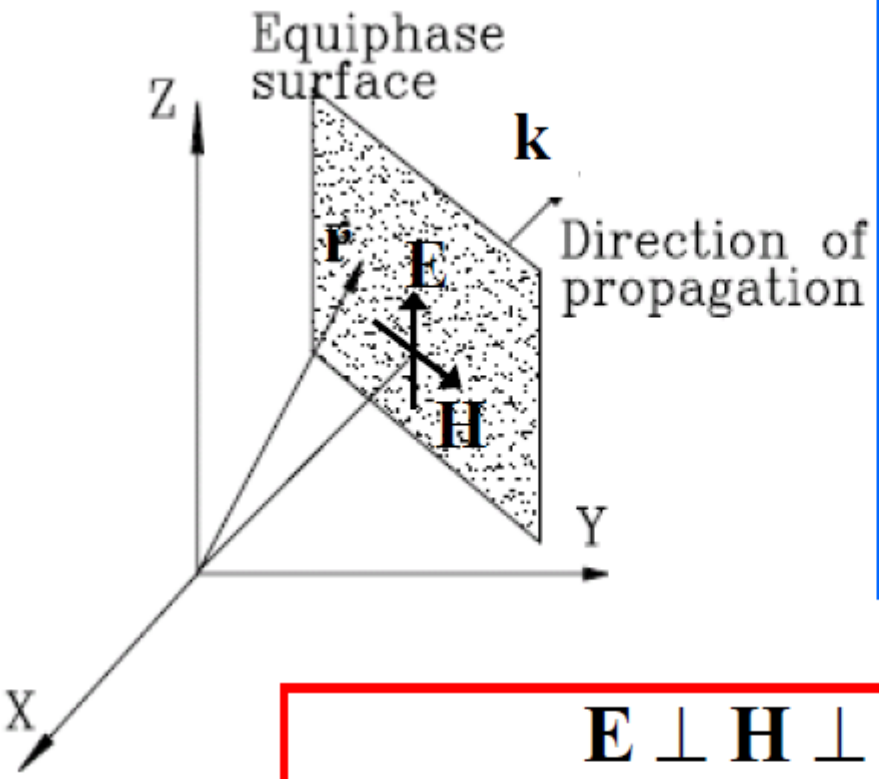
- The plane wave solutions represent electromagnetic waves which have the following properties:
  - The phase is constant on a plane which is perpendicular to direction of propagation
  - Direction of E-field and H-field are on the plane of constant phase.
  - There is no E-field or H-field in direction of propagation.

Plane Wave (cont.)

The plane wave is also called the **TEM** wave (TEM = Transverse ElectroMagnetic) in which $E_z = H_z = 0$ where z is the direction of propagation.

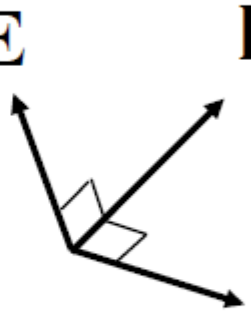


Plane Wave (cont.)



Right-hand rule:

(index finger) (thumb)



(middle finger)

$$\mathbf{E} \perp \mathbf{H} \perp \mathbf{k}$$

$$\mathbf{H} = \frac{1}{\eta}\hat{\mathbf{k}} \times \mathbf{E}, \qquad \mathbf{E} = \eta \mathbf{H} \times \hat{\mathbf{k}}$$

Transverse Electromagnetic Waves

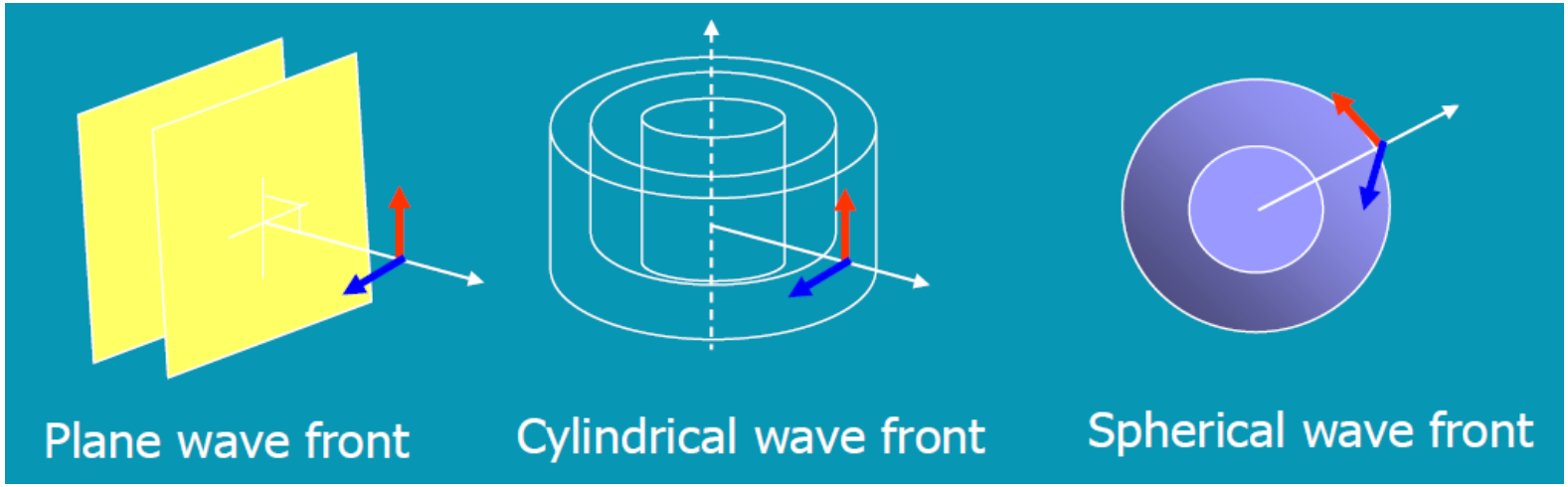
- The transverse electromagnetic (TEM) waves are the electromagnetic waves having the electric and magnetic fields perpendicular to the propagation direction.
- The general form of a TEM wave is

$$\mathbf{E} = \mathbf{E}_0 e^{-j\mathbf{k}\cdot\mathbf{r}}$$

$$\mathbf{H} = \mathbf{H}_0 e^{-j\mathbf{k}\cdot\mathbf{r}}$$

$$\mathbf{k} = k_x\hat{\mathbf{x}} + k_y\hat{\mathbf{y}} + k_z\hat{\mathbf{z}}$$

$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$



Instantaneous (Time-Domain) Fields

- Phasor form:

$$\mathbf{E}(z) = \mathbf{E}_0 e^{-jkz} = \hat{\mathbf{e}} E_0 e^{j\varphi} e^{-jkz}$$

$$\mathbf{H}(z) = \mathbf{H}_0 e^{-jkz} = \hat{\mathbf{h}} H_0 e^{j\varphi} e^{-jkz}$$

- Instantaneous solution:

$$\begin{aligned}
\mathbf{E}(z,t) &= \mathrm{Re}\left[\mathbf{E}(z) e^{j\omega t}\right] = \mathrm{Re}\left[\hat{\mathbf{e}} E_0 e^{j\varphi} e^{-jkz} e^{j\omega t}\right] \\
&= \hat{\mathbf{e}} E_0 \,\mathrm{Re}\left[e^{j(\omega t - kz + \varphi)}\right] = \hat{\mathbf{e}} E_0 \cos(\omega t - kz + \varphi) \\
\mathbf{H}(z,t) &= \mathrm{Re}\left[\mathbf{H}(z) e^{j\omega t}\right] = \mathrm{Re}\left[\hat{\mathbf{h}} H_0 e^{j\varphi} e^{-jkz} e^{j\omega t}\right] \\
&= \hat{\mathbf{h}} H_0 \,\mathrm{Re}\left[e^{j(\omega t - kz + \varphi)}\right] = \hat{\mathbf{h}} H_0 \cos(\omega t - kz + \varphi)
\end{aligned}$$

Phase Velocity

- The phase velocity is the propagation speed of a constant phase point on the wave.

$$\mathbf{E}(z,t) = \hat{\mathbf{e}} E_0 \cos(\omega t - kz + \varphi)$$

$$\psi(z,t) = \omega t - kz + \varphi = \text{constant}$$

$$\frac{\partial \psi}{\partial t} = \omega - k\frac{\partial z}{\partial t} = 0$$

$$u_p = \frac{\partial z}{\partial t} = \frac{\omega}{k} = \frac{\omega}{\omega\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu\varepsilon}}$$

Example

The electric field intensity of a 300-MHz uniform plane wave in free space is given as

$$\vec{E} = (20 + j50)(\vec{a}_x + 2\vec{a}_y)e^{jk_0 z} \quad \text{(V/m)}$$

- The propagation direction is $-z$.
- The wavelength is

$$\lambda_0 = \frac{c}{f} = \frac{3\times 10^8}{300\times 10^6} = 1\ \text{m}$$

- The phase velocity is

$$u_p = \frac{\omega}{k_0} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = c = 3\times 10^8\ \text{m/s}$$

Example (cont.)

- The propagation constant is

$$k_0 = \frac{2\pi}{\lambda_0} = 2\pi \text{ rad/m}$$

- The intrinsic impedance is

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi \approx 377\ \Omega$$

- The magnitude of the electric field is

$$\begin{aligned} \left|\vec{E}\right| &= \left|(20 + j50)(\vec{a}_x + 2\vec{a}_y)\right| \\ &= \sqrt{20^2 + 50^2} \times \sqrt{1^2 + 2^2} \\ &= 120.42 \quad (\text{V/m}) \end{aligned}$$

Example (cont.)

- The instantaneous (time-domain) electric field is

$$\begin{aligned}\vec{\mathbf{E}}(z,t) &= \mathrm{Re}\left[\vec{E}(z)e^{j\omega t}\right] = \mathrm{Re}\left[(20+j50)(\vec{a}_x+2\vec{a}_y)e^{jk_0 z}e^{j\omega t}\right] \\ &= (\vec{a}_x+2\vec{a}_y)\,\mathrm{Re}\left[53.85e^{j1.19}e^{jk_0 z}e^{j\omega t}\right] \\ &= (\vec{a}_x+2\vec{a}_y)53.85\cos(\omega t+k_0 z+1.19) \\ &= (\vec{a}_x+2\vec{a}_y)53.85\cos(2\pi(3\times10^8 t+z)+1.19)\ (\mathrm{V/m})\end{aligned}$$

At $t = 1$ ns, $z = 10$ cm:

$$\begin{aligned}\vec{\mathbf{E}}(10\ \mathrm{cm},1\ \mathrm{ns}) &= (\vec{a}_x+2\vec{a}_y)53.85\cos(2\pi(3\times10^8\times10^{-9}+0.1)+1.19) \\ &= (\vec{a}_x+2\vec{a}_y)53.85\cos(0.8\pi+1.19) \\ &= (\vec{a}_x+2\vec{a}_y)(53.85)(-0.85) = -(\vec{a}_x+2\vec{a}_y)45.77\ (\mathrm{V/m})\end{aligned}$$

Example (cont.)

- The phasor of the corresponding magnetic field is

$$\vec{H} = \frac{1}{\eta_0}\hat{k}\times\vec{E} = \frac{1}{120\pi}(-\vec{a}_z)\times\left[(\vec{a}_x + 2\vec{a}_y)(20+j50)e^{jk_0 z}\right]$$

$$= (2\vec{a}_x - \vec{a}_y)\frac{(20+j50)}{120\pi}e^{jk_0 z} \quad (\text{A/m})$$

- The instantaneous magnetic field is

$$\vec{\mathbf{H}}(z,t) = \frac{1}{\eta_0}\hat{k}\times\vec{\mathbf{E}}(z,t) = \text{Re}\left[\vec{H}(z)e^{j\omega t}\right]$$

$$= (2\vec{a}_x - \vec{a}_y)\frac{53.85}{120\pi}\cos(2\pi(3\times10^8 t + z) + 1.19) \quad (\text{A/m})$$

Example

The instantaneous expression for the magnetic field intensity of a uniform plane wave propagating in y-direction in air is given by

$$\vec{H} = \vec{a}_z 2\times 10^{-6} \cos(10^8 \pi t - k_0 y + 1.5) \quad (\text{A/m})$$

a) Determine the value of k_o.
b) Write the instantaneous expression for the electric field.

Example (cont.)

$$\vec{H} = \vec{a}_z 2\times10^{-6} \cos(10^8 \pi t - k_0 y + 1.5) \quad (\text{A/m})$$

- $\vec{a}_z$: direction
- 2×10^{-6}: amplitude
- $10^8 \pi t$: ωt
- $k_0 y$: Phase constant
- 1.5: Phase reference at t = 0

Phase constant of plane wave in lossless medium

$$k_0 = \frac{\omega\sqrt{\varepsilon_r \mu_r}}{3\times10^8} = \frac{10^8 \pi \sqrt{1\times1}}{3\times10^8} = \frac{\pi}{3} \quad (\text{rad/m})$$

Example (cont.)

$$\vec{H} = \vec{a}_z 2\times 10^{-6} \cos(10^8 \pi t - k_0 y + 1.5) \quad \text{(A/m)}$$

Change to phasor form

$$\vec{H} = \vec{a}_z 2\times 10^{-6} e^{j1.5} e^{-jk_0 y}$$

compare with standard form of phase term

$$e^{j\vec{k}\cdot\vec{r}} = e^{-j(k_x x + k_y y + k_z z)} = e^{-jk_0 y}$$

Hence, $k_x = 0, k_y = k_0, k_z = 0$

$$\vec{k} = k_0 \vec{a}_y$$

Example (cont.)

$$\vec{k}\cdot\vec{E}=k_y E_y=0\rightarrow E_y=0$$

$$\vec{k}\times\vec{E}=\begin{vmatrix}\vec{a}_x & \vec{a}_y & \vec{a}_z\\ 0 & k_y & 0\\ E_x & 0 & E_z\end{vmatrix}=k_y E_z\vec{a}_x-k_y E_x\vec{a}_z=\vec{a}_z\omega\mu\left|\vec{H}\right|$$

$$\text{Hence,}\quad E_z=0\ \text{ and }\ -k_y E_x=\omega\mu\left|\vec{H}\right|$$

$$\vec{E}=E_x\vec{a}_x=-\frac{\omega\mu}{k_0}\left|\vec{H}\right|\vec{a}_x=-\sqrt{\frac{\mu_r\mu_0}{\varepsilon_r\varepsilon_0}}\left|\vec{H}\right|\vec{a}_x\ \ (\text{V/m})$$

Phasor $$\vec{E}=-\vec{a}_x\left(377\times 2\times 10^{-6}e^{j1.5}e^{-jk_o y}\right)\ \ (\text{V/m})$$

Time-domain $$\vec{E}=-\vec{a}_x\left(377\times 2\times 10^{-6}\cos(2\pi\times 10^8 t-k_o y+1.5)\right)\ \ (\text{V/m})$$

Polarization

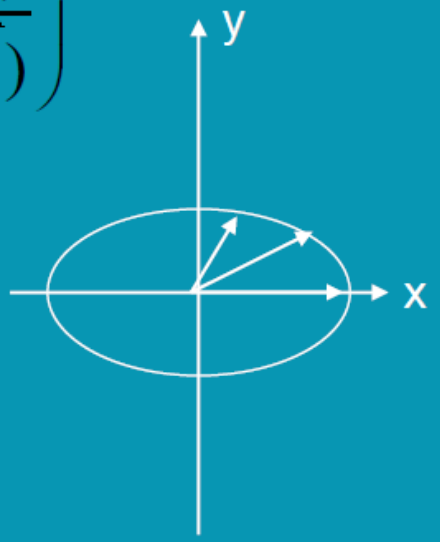
Determine magnitude of E-field

$$\vec{E} = \vec{a}_x 2\cos(10^8 t - z/\sqrt{3}) - \vec{a}_y \sin(10^8 t - z/\sqrt{3})$$

$$\theta = \tan^{-1}\frac{E_y}{E_x} = \tan^{-1}\left(\frac{-\sin(10^8 t - z/\sqrt{3})}{2\cos(10^8 t - z/\sqrt{3})}\right)$$

$$\theta = \tan^{-1}\left(\frac{1}{2}\tan(10^8 t - z/\sqrt{3})\right)$$

θ varies with time!.



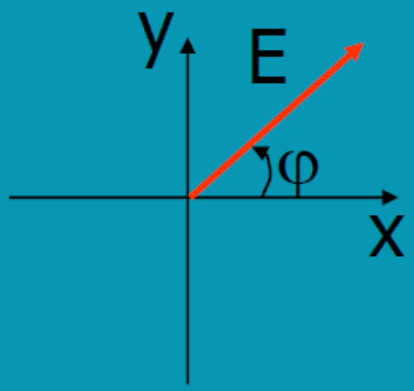
Polarization (cont.)

Let us write component of E-field as

$$E_x = C_x \cos(\omega t + \beta z)$$

$$E_y = C_y \cos(\omega t + \beta z + \varphi)$$

At plane z = 0, the magnitude and direction of E-field become



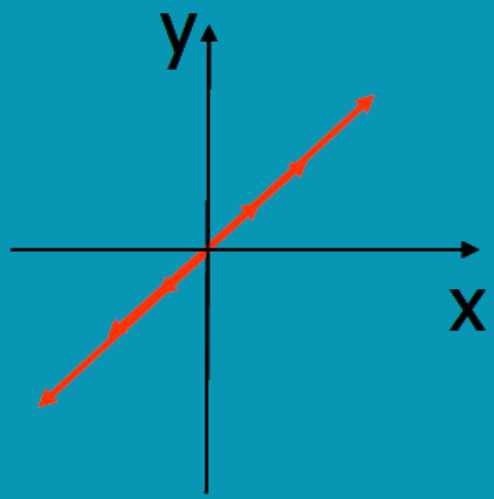
$$\left|\vec{E}\right|^2 = C_x^2 \cos^2(\omega t) + C_y^2 \cos^2(\omega t + \varphi)$$

$$\phi = \tan^{-1} \frac{C_y \cos(\omega t + \varphi)}{C_x \cos(\omega t)}$$

Polarization (cont.)

Consider when φ =0 (Linear Polarization)

At plane z = 0, the magnitude of E-field is



$$\left|\vec{E}\right|^2 = C_x^2 \cos^2(\omega t) + C_y^2 \cos^2(\omega t)$$

$$\left|\vec{E}\right| = \sqrt{C_x^2 + C_y^2}\cos(\omega t)$$

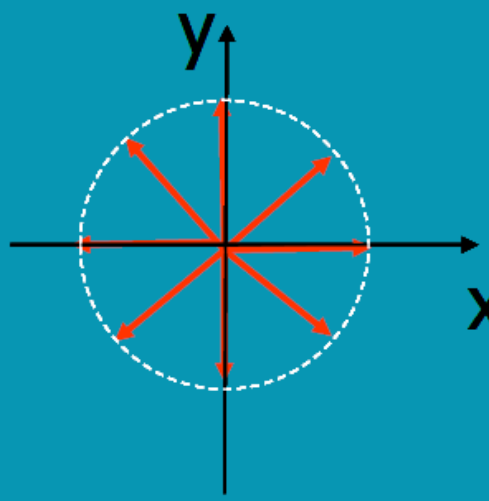
The magnitude of E-field is oscillating with angular frequency ω.

$$\phi = \tan^{-1}\frac{C_y}{C_x}$$

Polarization (cont.)

Consider when $\varphi = -\pi/2$ and $C_x = C_y$ (Right Hand Circular Polarization)

At plane z = 0, the magnitude and direction of E-field is



$$\left|\vec{E}\right|^2 = C_x^2 \cos^2(\omega t) + C_y^2 \cos^2(\omega t - \frac{\pi}{2})$$

$$C_y = C_x$$

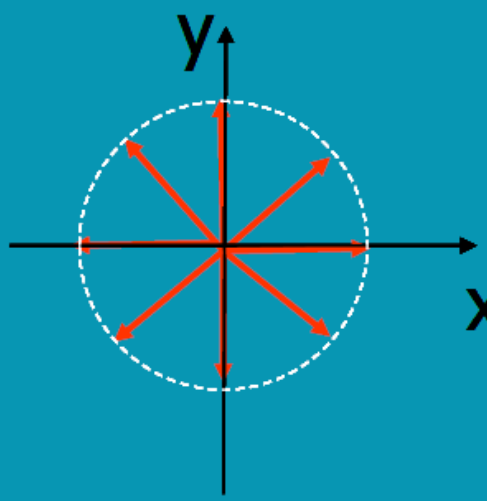
$$\left|\vec{E}\right|^2 = C_x^2 \cos^2(\omega t) + C_x^2 \sin^2(\omega t) = C_x^2$$

$$\phi = \tan^{-1} \frac{C_x \sin(\omega t)}{C_x \cos(\omega t)} = \tan^{-1} \tan(\omega t) = \omega t$$

Polarization (cont.)

Consider when $\varphi = \pi/2$ and $C_x = C_y$ (Left Hand Circular Polarization)

At plane z = 0, the magnitude and direction of E-field is



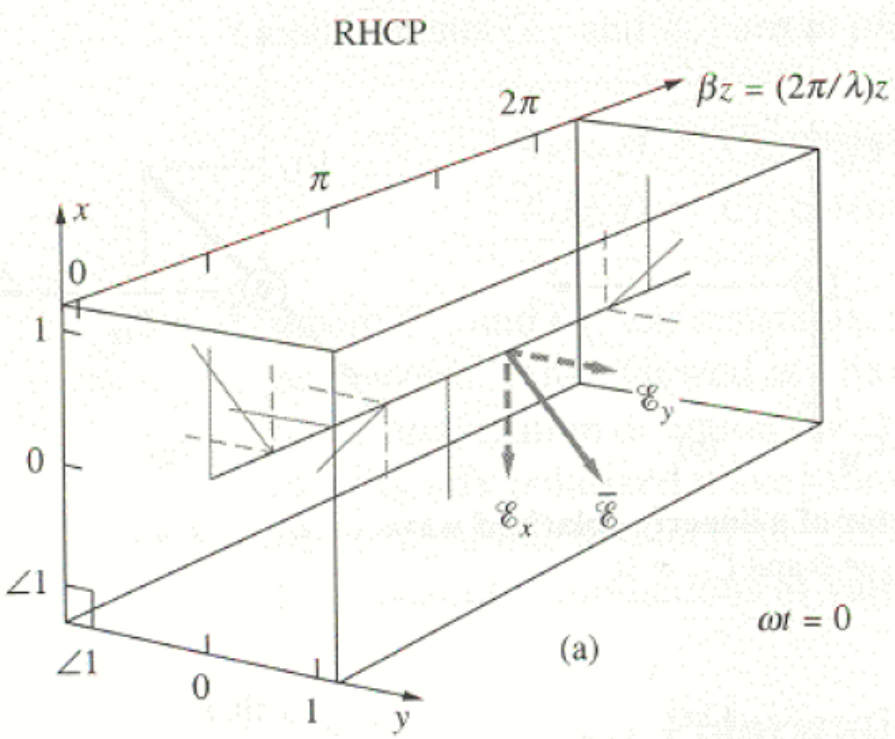
$$\left|\vec{E}\right|^2 = C_x^2 \cos^2(\omega t) + C_y^2 \cos^2(\omega t + \frac{\pi}{2})$$

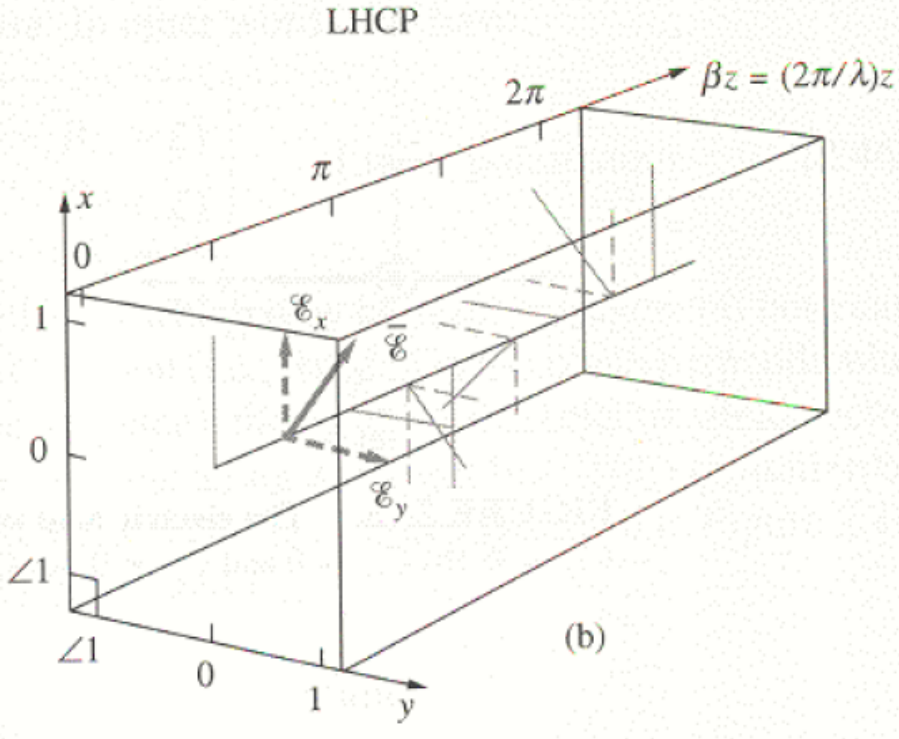
$$C_y = C_x$$

$$\left|\vec{E}\right|^2 = C_x^2 \cos^2(\omega t) + C_x^2 \sin^2(\omega t) = C_x^2$$

$$\phi = \tan^{-1} \frac{-C_x \sin(\omega t)}{C_x \cos(\omega t)} = -\tan^{-1} \tan(\omega t) = -\omega t$$

Polarization (cont.)





Polarization (cont.)

Consider when $\varphi = -\pi/2$ and $C_x > C_y$ (Right Hand Circular Polarization)

At plane z = 0, the magnitude of E-field is

$$\cos(\omega t) = \frac{E_x(0,t)}{C_x};$$

$$\sin(\omega t) = \frac{E_y(0,t)}{C_y}$$

$$\sin(\omega t) = \sqrt{1-\cos^2(\omega t)}$$

$$\sin(\omega t) = \sqrt{1-\frac{E_x^2(0,t)}{C_x^2}}$$

$$\left|\vec{E}\right|^2 = C_x^2\cos^2(\omega t) + C_y^2\cos^2(\omega t - \frac{\pi}{2})$$

$$\left|\vec{E}\right|^2 = C_x^2\cos^2(\omega t) + C_y^2\sin^2(\omega t)$$

$$\frac{E_y(0,t)}{C_y} = \sqrt{1-\frac{E_x^2(0,t)}{C_x^2}}$$

$$\frac{E_x^2(0,t)}{C_x^2} + \frac{E_y^2(0,t)}{C_y^2} = 1$$

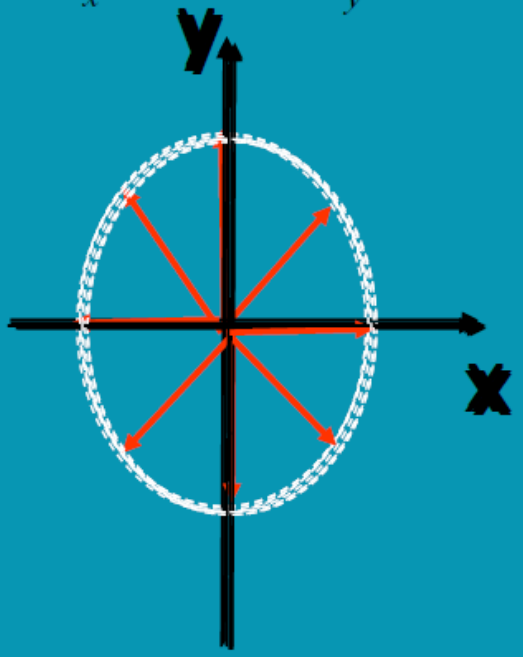
Equation of ellipse:

Polarization (cont.)

Consider when $\varphi = -\pi/2$ and $C_x > C_y$ (Right Hand Circular Polarization)

At plane z = 0, the magnitude of E-field is

$$\frac{E_x^2(0,t)}{C_x^2} + \frac{E_y^2(0,t)}{C_y^2} = 1$$



Equation of ellipse:

$$\varphi = \tan^{-1}\frac{C_y \sin(\omega t)}{C_x \cos(\omega t)}$$

$$\varphi = \tan^{-1}\frac{C_y}{C_x}\tan(\omega t)$$

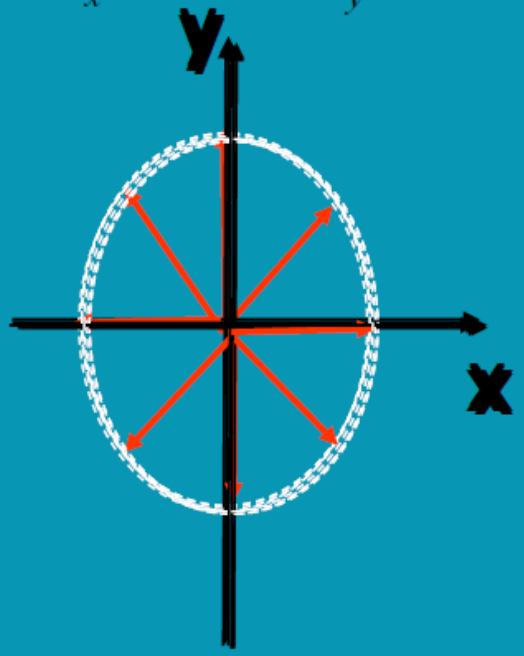
$$\tan\varphi = \frac{C_y}{C_x}\tan(\omega t)$$

Polarization (cont.)

Consider when $\varphi = \pi/2$ and $C_x > C_y$ (Right Hand Circular Polarization)

At plane z = 0, the magnitude of E-field is

$$\frac{E_x^2(0,t)}{C_x^2} + \frac{E_y^2(0,t)}{C_y^2} = 1$$



Equation of ellipse:

$$\varphi = \tan^{-1}\frac{-C_y \sin(\omega t)}{C_x \cos(\omega t)}$$

$$\varphi = \tan^{-1}\frac{-C_y}{C_x}\tan(\omega t)$$

$$\tan\varphi = -\frac{C_y}{C_x}\tan(\omega t)$$